

Worked for a couple of seconds

Yes. Based on the **proposed features**, I would now move from the high-level system structure to a proper **Functional Requirements & Role-Permission Matrix**. The proposed document specifically recommends expanding the original brief into a detailed requirements and architecture specification before full coding.

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Below is the structure I recommend for the **actual LisheBora system specification**.

LisheBora e-Sourcing Platform

Functional Requirements & System Structure

1. System Architecture

The platform should be organized into **14 business domains**, rather than treating every feature as a separate screen. This follows the proposed architecture in the review document.

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#	Domain	Main Purpose
1	Identity & Access	Users, roles, permissions and security
2	Master Data	Counties, schools, commodities, units, terms and configurations
3	Demand & Nutrition	School demand, menus, feeding requirements and nutrition
4	Budget Management	Budgets, commitments, balances and controls
5	Supplier Management	Registration, prequalification and Supplier 360°
6	Procurement	RFQs, bids, evaluation, awards and POs
7	Contract Management	Contracts, framework agreements and call-offs
8	Aggregation & Quality	Farmer aggregation, batches and quality inspections
9	Inventory & Warehousing	Stock receiving, movement, allocation and reconciliation

#	Domain	Main Purpose
10	Logistics & Delivery	Dispatch, transport, delivery and POD
11	Finance	Invoices, matching, approvals and payment tracking
12	Complaints & Performance	Grievances and supplier performance
13	MEAL & Analytics	Dashboards, indicators, reports and analytics
14	Platform Services	Workflow, notifications, documents, audit and integrations

The proposed end-to-end flow is:

Demand → Budget → Sourcing → Evaluation → Contract/PO → Aggregation → Quality → Inventory → Logistics → Delivery → Invoice → Payment → Performance → MEAL.

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2. Identity & Access Management

Purpose

Control who can access the platform and what they are allowed to do.

Main features

- User registration
- Login/logout
- Password management
- Role assignment
- Organization assignment
- County/school-level access
- Permission management
- MFA
- Account activation/deactivation
- Session management
- Password reset
- Login history
- Audit trail

Important design principle

Do **not** simply say:

"This user is a Procurement Officer."

Instead, the system should understand:

User + Role + Organization + Geographic Scope + Permission

For example:

John

- Role: Procurement Officer
- County: Kisumu
- Permissions: Create RFQ, View Bids, Submit Evaluation
- Cannot: Approve Award, Make Payment

This supports segregation of duties.

The proposed design explicitly requires granular permissions such as create, view, edit, submit, verify, evaluate, approve, reject, pay, export and audit.

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3. Master Data Management

This should be the foundation of the entire system.

Main entities

Administrative

- County
- Sub-county
- Ward
- Location

Education

- School
- School type
- School cluster
- Enrolment
- Academic year
- Term
- Feeding days

Procurement

- Commodity
- Commodity category
- Unit of measure
- Procurement method
- Procurement thresholds
- Procurement criteria

Nutrition

- Menu
- Food portion
- Nutrition standard
- Commodity specification

Supply chain

- Aggregation centre
- Warehouse
- Supplier category
- Quality standard

Configuration

Administrators should be able to configure:

- Procurement thresholds
- Approval thresholds
- Feeding days
- Commodity specifications
- Quality standards
- Notification rules
- Workflow rules
- Document types
- User permissions

This is important because these rules may change without requiring developers to modify the entire system.

4. School & Demand Management

This is where procurement begins.

School profile

Each school should have:

- School name
- School code
- County
- Sub-county
- Location
- GPS coordinates
- Enrolment
- Feeding programme
- Storage capacity
- Contact person
- Bank/payment information where required
- Active/inactive status

Demand planning

School officer enters:

- Number of learners
- Expected attendance
- Feeding days
- Menu
- Existing stock
- Required commodities
- Available budget

The system calculates:

Learners × Feeding Days × Standard Portion = Required Quantity

Then subtract:

Required Quantity – Available Stock = Procurement Requirement

Consolidation

The system should consolidate demand:

School → Cluster → Sub-county → County

This creates procurement opportunities.

5. Menu & Nutrition

The system should allow authorized users to define:

- Weekly menus
- Term menus
- Commodity quantities
- Portion sizes
- Nutrition requirements
- Approved alternatives
- Seasonal alternatives

The system should then show:

- Calories/nutrition indicators
- Commodity diversity
- Menu compliance
- Demand generated from the menu

6. Budget Management

This is one of the important additions from the proposed features.

The system should maintain:

Budget

- Funding source
- County
- School
- Financial year
- Budget amount

Budget status

Approved Budget



Committed



Spent



Balance

For example:

Budget	Amount
Approved	KSh 10,000,000

Budget	Amount
Committed	KSh 6,000,000
Spent	KSh 4,000,000
Available	KSh 4,000,000

Controls

Before issuing a PO:

System checks available budget.

If insufficient:

"Insufficient budget. Approval required for exception."

The proposed review specifically identifies budget commitment control, budget alerts, overrun exceptions and utilization reporting as missing areas that should be added.

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7. Supplier Management — Supplier 360°

This should be much more than supplier registration.

Supplier profile

- Business name
- Registration number
- Supplier type
- Contact details
- Location
- Ownership
- Gender/age/PWD inclusion information
- Farmer group membership
- Commodities supplied
- Capacity
- Storage
- Transport capacity
- Payment information

Documents

- Registration certificate
- Tax documents
- Food safety certificates
- Licences
- Insurance
- Other required documents

Supplier 360°

The system should show the complete supplier history:

Profile → Documents → Bids → Awards → Contracts → Deliveries → Quality → Invoices → Payments → Complaints → Performance

This is one of the strongest proposed enhancements in the review.

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8. E-Procurement

This is the core e-sourcing functionality.

Procurement plan

- Procurement requirement
- Commodity
- Quantity
- Estimated value
- Budget
- Procurement method
- Timeline

Sourcing event

Create:

- RFQ
- Competitive bidding
- Framework agreement
- Call-off
- Other permitted procurement method

RFQ fields

- RFQ number
- Title

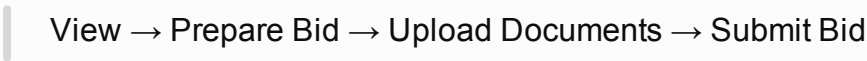
- Commodity
- Quantity
- Specification
- Lot
- Delivery location
- Closing date
- Evaluation criteria
- Required documents
- Terms and conditions

Supplier side

Supplier sees:

Available Opportunities

Then:



Once submitted, the bid should become locked after the deadline.

The proposed workflow explicitly includes secure bid locking at closing.

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9. Bid Evaluation & Approval

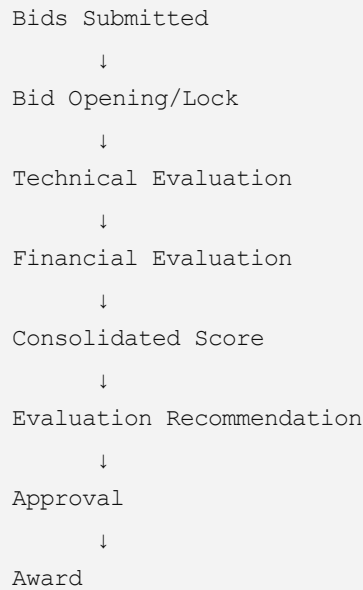
Evaluation structure

Example:

Criterion	Weight
Price	40%
Technical compliance	20%
Capacity	15%
Quality	10%
Local sourcing	10%
Inclusion	5%

The exact criteria should remain configurable.

Workflow



Conflict of interest

Evaluation committee members should declare conflicts before evaluating.

10. Contract Management

This is another major addition.

Contract

- Contract number
- Supplier
- Commodity
- Contract value
- Contract quantity
- Start date
- End date
- Delivery schedule
- Payment terms
- Contract status

Contract balance

The system should track:

Contract Quantity

minus

Quantity Already Ordered

equals

Remaining Quantity

And similarly:

Contract Value – Amount Used = Remaining Contract Value

Additional functions

- Amendments
- Versions
- Call-offs
- Expiry alerts
- Performance
- Contract history

The proposed review specifically recommends contract versioning, quantity/value balances, schedules, call-offs, amendments and expiry alerts.

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11. Aggregation & Quality

Once an order is awarded, the system moves into fulfilment.

Aggregation

Capture:

- Farmer/supplier
- Commodity
- Variety
- Quantity
- Date
- Location
- Aggregation centre
- Batch number

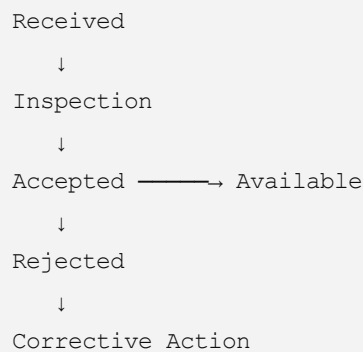
Quality inspection

Inspector records:

- Commodity

- Batch
- Quantity
- Grade
- Quality parameters
- Inspection result
- Photos
- Test results
- Accepted quantity
- Rejected quantity
- Rejection reason
- Corrective action

Status



12. Inventory & Warehouse

This should be a full module rather than simply recording school stock.

Warehouse

- Warehouse registration
- Location
- Capacity
- Responsible officer

Stock transactions

- Goods received
- Stock movement
- Reservation
- Allocation
- Transfer
- Adjustment

- Damage
- Rejection
- Waste
- Stock count
- Reconciliation

Stock status

```
Opening Stock
+ Receipts
+ Transfers In
- Allocations
- Dispatches
- Waste
- Adjustments
= Closing Stock
```

Alerts

- Low stock
- Expiring stock
- Expired stock
- Unusual adjustments

These inventory capabilities were specifically identified as a gap in the original brief.

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13. Logistics & Delivery

Dispatch

Create:

- Dispatch number
- PO
- Supplier
- Warehouse
- Destination school
- Commodity
- Quantity
- Batch
- Vehicle
- Driver
- Dispatch date

Delivery tracking

Planned

↓

Dispatched

↓

In Transit

↓

Delivered

↓

School Verification

↓

Accepted / Partially Accepted / Rejected

Proof of delivery

Capture:

- Quantity received
- Condition
- Date/time
- Receiver
- Signature
- Photo
- Remarks

14. Finance

The system should implement the proposed **three-way matching**.

Three documents

Purchase Order

-

Goods Receipt

-

Invoice

↓

Three-way match

If they agree:

Invoice can proceed for approval.

If they do not:

Exception generated.

Invoice workflow

```
graph TD; A[Invoice Submitted] --> B[Validation]; B --> C[3-Way Match]; C --> D[Finance Review]; D --> E[Approval]; E --> F[Payment]; F --> G[Paid];
```

The system should track payment status even where the actual payment happens in an external finance or banking system.

15. Workflow & Approval Engine

This should be a **central platform service**, not something independently coded into every module.

For example:

```
Demand Approval
Budget Approval
RFQ Approval
Evaluation Approval
Award Approval
PO Approval
Invoice Approval
Exception Approval
```

Each can use configurable:

- Approval stages
- Thresholds
- Approvers

- Delegates
- Escalation
- SLA
- Rejection
- Resubmission

The proposed review specifically recommends a reusable configurable workflow engine.

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16. Risk & Exception Management

Create a central **Exceptions Centre**.

Examples:

- Late supplier
- Late delivery
- Quantity shortage
- Quality rejection
- Expired certificate
- Budget overrun
- Invoice mismatch
- Contract expiry
- Repeated complaints

Each exception should have:

- Reference
- Category
- Severity
- Related transaction
- Responsible officer
- Date
- Action required
- Corrective action
- Due date
- Status
- Resolution

The proposed system specifically calls for risk flags, exception workflows, corrective actions and risk dashboards.

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17. QR / Barcode Traceability

This should eventually allow:

Farmer → Aggregator → Batch → Warehouse → Vehicle → School

Example:

BATCH-2026-KSM-000145

At every stage:

Scan QR → Update transaction

The system can therefore answer:

Where did this commodity come from?

Which school received it?

Which supplier supplied it?

Which batch was delivered?

What was the quality result?

The proposed feature specifically includes QR/barcode generation and scanning at aggregation, dispatch and receipt.

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18. Offline Capability

This should be designed carefully rather than simply saying "offline enabled."

Offline process

User captures data

↓

Local storage

↓

Transaction queued

↓

Internet available

↓

Automatic synchronization

↓

Server validation



Success / Conflict / Error

The system should support:

- Offline forms
- Offline photos
- Queued transactions
- Sync status
- Retry
- Conflict detection
- Conflict resolution

These requirements are explicitly proposed in the review.

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19. GIS

Later phase but the database should be designed to support it.

Map:

- Schools
- Suppliers
- Farmer groups
- Aggregation centres
- Warehouses
- Delivery locations

Possible analysis:

Supplier → School distance

Supply gaps by location

Delivery routes

Geographic procurement patterns

The proposed technical architecture recommends PostgreSQL with PostGIS if GIS capabilities are adopted.

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20. Notifications

One central notification service should serve the entire system.

Channels

- In-app
- SMS
- Email
- Push notification

Events

Examples:

RFQ published

RFQ closing tomorrow

Bid submitted

Approval required

Certificate expiring

PO approved

Delivery delayed

Invoice rejected

Payment completed

21. Document Management

Every important document should have a central document record.

Document metadata

- Document type
- Owner
- Related entity
- Version
- Issue date
- Expiry date

- Status
- Access permissions

Documents can be linked to:

- Supplier
- RFQ
- Bid
- Contract
- PO
- Inspection
- Delivery
- Invoice

The proposed document management service includes versioning, expiry, access controls, transaction linking, retention and audit.

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22. Complaints & Grievance Management

Users should be able to report:

- Procurement complaint
- Supplier complaint
- Delivery issue
- Quality issue
- Payment issue
- School issue
- System issue

Case workflow

```
graph TD; Submitted --> Assigned; Assigned --> Investigation; Investigation --> Action; Action --> Resolution; Resolution --> Closed;
```

Submitted

↓

Assigned

↓

Investigation

↓

Action

↓

Resolution

↓

Closed

Every case should have an audit history.

23. Supplier Performance

The system should automatically build the supplier scorecard from transactions.

Possible indicators:

- On-time delivery
- Complete delivery
- Quality acceptance
- Rejection rate
- Contract fulfilment
- Response time
- Complaints
- Invoice/payment history

Rather than asking officers to manually calculate performance, the system should calculate it from actual transactions.

24. MEAL & Management Dashboard

Executive dashboard

Show:

- Total procurement value
- Procurement by county
- Procurement by commodity
- Supplier participation
- Local sourcing
- Women/youth/PWD participation
- Delivery performance
- Payment performance
- Quality rejection
- Complaints
- Budget utilization

Operational dashboard

For procurement officers:

- Pending RFQs
- Bids awaiting evaluation
- Approvals pending
- Expiring contracts

- Supplier documents expiring
- Delayed deliveries

School dashboard

- Demand
 - Orders
 - Deliveries
 - Stock
 - Menu
 - Complaints
-

25. Advanced Analytics — Later Phase

The architecture should allow future:

- Demand forecasting
- Supply forecasting
- Price intelligence
- Supplier analytics
- Procurement anomaly detection
- Predictive alerts
- Forecast accuracy

The review recommends these as later/advanced capabilities rather than making them part of the initial MVP.

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26. Recommended User Roles

I would use the following starting list:

1. Super Administrator
2. System Administrator
3. County Administrator
4. County Procurement Officer
5. County Finance Officer
6. School Administrator
7. School Meals Officer
8. Nutrition Officer
9. Procurement Officer
10. Evaluation Committee Member
11. Approving Officer

12. Supplier
13. Farmer Group/Cooperative
14. Aggregator
15. Warehouse Officer
16. Quality Inspector
17. Logistics Officer
18. Driver
19. Finance Officer
20. Programme/MEAL Officer
21. Risk/Compliance Officer
22. Auditor
23. Helpdesk Officer

The source review explicitly proposes most of these roles and requires granular permissions supporting segregation of duties.

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27. Permission Model

Instead of hard-coding permissions into individual pages, create a central permission engine.

Actions

Permission	Meaning
VIEW	View information
CREATE	Create record
EDIT	Modify record
SUBMIT	Submit for workflow
VERIFY	Verify information
EVALUATE	Evaluate
APPROVE	Approve
REJECT	Reject
PAY	Process/authorize payment

Permission	Meaning
EXPORT	Export information
AUDIT	View audit information

Example:

Role	RFQ	Bid	Evaluation	Award	Payment
Procurement Officer	Create/Edit	View	Prepare	Submit	No
Supplier	View/Respond	Create	No	No	View
Evaluator	View	View	Evaluate	Recommend	No
Approving Officer	View	View	View	Approve	No
Finance Officer	View	View	View	View	Process
Auditor	View	View	View	View	Audit

This is the type of matrix I would build next.

28. Technical Architecture

The proposed technology stack is:

Frontend

- React
- Next.js
- TypeScript
- Responsive/PWA

Backend

- Python
- FastAPI
- Pydantic
- SQLAlchemy

Database

- PostgreSQL
- PostGIS if GIS is implemented

File storage

- S3-compatible object storage

Background processing

- Redis
- Task/queue service

API

- Versioned REST API
- `/api/v1/`

Authentication

- OAuth2/OIDC
- RBAC
- Optional MFA

Deployment

- Containers
- Development
- Staging
- Production

Monitoring

- Centralized logs
- Error monitoring
- Infrastructure metrics
- Alerts

These are directly recommended in the proposed technical architecture.

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29. Development Roadmap

I would **not develop everything at once**.

Phase 0 — Requirements & Design

Produce:

- BRD
- SRS

- Process maps
- ERD
- UI prototype
- Role-permission matrix
- Integration specification

The source review recommends this as the first phase.

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Phase 1 — Foundation

- Authentication
- RBAC
- Master data
- Schools
- Suppliers
- Documents
- Audit trail
- Configuration

Phase 2 — Demand & Budget

- School demand
- Menu
- Stock
- Demand calculation
- Consolidation
- Procurement planning
- Budget control

Phase 3 — E-Procurement

- RFQs
- Bids
- Evaluation
- Approval
- Awards
- POs
- Contracts

Phase 4 — Supply Chain

- Aggregation
- Quality
- Batches
- Inventory

- Dispatch
- Logistics
- Delivery

Phase 5 — Finance & Performance

- Invoices
- Three-way matching
- Payment tracking
- Reconciliation
- Supplier performance
- Complaints
- MEAL

Phase 6 — Advanced

- Offline sync
- GIS
- QR
- Integrations
- Forecasting
- Advanced analytics
- Optimization

This sequencing is essentially the development approach recommended in the proposed document.

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30. What I recommend we build next

The **next document should not yet be the database or code.**

We should first create a proper:

LisheBora Functional Requirements Specification (FRS)

For **every module**, it should contain:

1. Module
2. Sub-module
3. Feature
4. User
5. User action
6. Screen
7. Fields

8. Validation rules
9. Business rules
10. Workflow
11. Approval requirements
12. Statuses
13. Notifications
14. Documents
15. Audit requirements
16. Reports
17. Permissions
18. Dependencies
19. Acceptance criteria

For example:

Supplier Registration → Create Supplier

would be specified down to the individual fields, validation, documents, approval process, notification, status changes and permissions.

That will give the developer something **much closer to a buildable blueprint**, rather than simply a list of modules. It also directly addresses the review's conclusion that the current brief needs to be expanded into a detailed System Requirements and Architecture Specification before full-scale coding.

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I would structure that FRS in the next step module-by-module, starting with Identity & Access, Master Data, School/Demand, Budget, Supplier 360°, and then moving through the complete procurement-to-payment workflow.

Sources